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## ASHWANI TAYAL

AVP / SOLUTION ARCHITECT / SOFTWARE ENGINEER

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### OBJECTIVE

- **Strategic Leadership:** Accomplished **Solution Architect & Engineering Leader** with 14+ years of experience in designing and scaling complex IT ecosystems.
- **Digital Transformation:** Proven expertise in delivering high-impact **Digital Governance** and **Enterprise solutions** for central and state government bodies.
- **Result-Oriented:** Seeking a leadership role to leverage deep technical roots in **Full-Stack Development** and **System Architecture** to drive organizational growth and innovation.

### PROFESSIONAL PROFILE

- **Engineering Leadership:** Strategic Technical Leader with **14+ years** of experience architecting and deploying large-scale digital governance and enterprise-grade software solutions.
- **Rapid Product Delivery:** Proven track record of taking complex systems from concept to launch under tight deadlines, recently delivering the **Divyang Sahayak Portal** for the Maharashtra Government in **less than 90 days**.
- **Technical Breadth:** Expert in building high-availability systems using **PHP (Yii2/Laravel)**, **Python (Django)**, and **Microservices architecture** integrated with cloud-native infrastructure (**AWS/GCP/Azure**).
- **Government & Public Sector Impact:** Delivered mission-critical platforms for the **Ministry of Health (MoHFW)**, **NCDC**, **Digital India**, and the **Government of Maharashtra**, impacting millions of citizens.
- **Architecture Specialist:** Deep expertise in **API-first design**, database optimization (**PostgreSQL/MongoDB**), and translating complex business requirements into scalable technical roadmaps (**HLD/LLD**).

### EDUCATION

- **Master of Technology (M. Tech) in Computer Science & Engineering** | 2014
- **School of PG Engineering, Maharshi Dayanand University (MDU)**
- **Bachelor of Technology (B. Tech) in Information Technology** | 2011
- **Laxmi Devi Institute of Engineering and Technology, RTU**



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## PERSONAL DETAILS

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## TECHNICAL SKILLS

### Architecture & Design

- **Methodologies:** System Design (HLD/LLD), Microservices Architecture, API-First Design, Scalable Web Architectures, Serverless patterns.
- **Frameworks:** Object-Oriented Programming (OOP), Model-View-Controller (MVC), Design Patterns, SDLC/Agile/Scrum.
- **Strategic Planning:** SRS/FRS Documentation, Technical Roadmaps, KPI Identification, Stakeholder Management, Vendor Coordination.

### Backend & Frameworks

- **Languages:** PHP (Yii2, Yii, Laravel, CodeIgniter), Python (Django), Java (J2EE/JSP).
- **API Ecosystems:** RESTful API Development, Third-party integrations (Payment Gateways, Etsy, Shopify, UMANG, IIHP, IVRS).

### Data & Infrastructure

- **Databases:** PostgreSQL (Advanced Query Optimization, Schema Design), MySQL, MongoDB.
- **Cloud & Platforms:** AWS (EC2, S3, RDS), Google Cloud Platform (GCP), Microsoft Azure.
- **Data Analytics:** Apache Superset (BI Dashboards), Data Migration, Bulk Data Processing (ETL).

### Frontend & Mobile

- **Web Technologies:** React, JavaScript (ES6+), jQuery, HTML5, CSS3.
- **Mobile Strategy:** Offline-first architecture, Data Synchronization (Sync-on-Online), Mobile-Web Bridge APIs.

### Operating Systems & Tools

- **Environment:** Linux (Ubuntu/CentOS), Windows Server.
- **DevTools:** VS Code, Git, Bitbucket, Mercurial, Vim, FileZilla.

## WORK EXPERIENCE

### Primus Partners

Assistant Vice President (October 2023 - Present)

### Projects:

#### Divyang Sahayak Portal (Maharashtra Government)

*Role: Solution Architect & Lead Backend Developer*

- **Rapid Development:** Architected and deployed a state-wide digital empowerment portal for Persons with Disabilities (PwD) in a record turnaround time of under 3 months.
- **Scalability & Impact:** Designed the system to facilitate seamless service delivery and welfare scheme access for the Divyang Welfare Department, inaugurated by the Chief Minister of Maharashtra.



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- **Technical Execution:** Developed a robust backend using PHP/Yii2 and PostgreSQL, ensuring high security and data integrity for sensitive citizen information.
- **Integration:** Implemented an API-driven architecture to ensure the portal could interface with existing government databases for real-time verification and processing.

### **Uttarakhand Parivar Pechan Patra (UKPPP)**

Role: Solution Architect

Technologies: PHP (Yii2), Python (Django), React, PostgreSQL, REST APIs

- **Architectural Strategy:** Designed and orchestrated a state-wide digital identity framework using a hybrid-stack architecture; leveraged Django for high-performance backend services and Yii2 for robust administrative modules.
- **System Interoperability:** Engineered a centralized API gateway to facilitate seamless, secure data exchange between multiple government departments, enabling unified service delivery for millions of families.
- **Data Integrity & Scalability:** Modelled a complex relational database schema in PostgreSQL to ensure data consistency across diverse government datasets while maintaining sub-second query performance.
- **End-to-End Leadership:** Managed the full product lifecycle (SDLC), from initial system design to stakeholder alignment, ensuring the platform met stringent state digital governance and security standards.
- **UI/UX Modernization:** Directed the development of a responsive, citizen-centric interface using React, significantly improving accessibility and user engagement for state residents.

### **Maharashtra Tree Plantation Monitoring System**

Role: Solution Architect & Lead Backend Developer

Technologies: PHP (Yii2), PostgreSQL, REST APIs, Mobile Integration

- **Scalable Architecture:** Architected a state-wide monitoring ecosystem to track large-scale plantation drives, ensuring system reliability across geographically dispersed regions.
- **Edge Data Management:** Engineered a robust offline-first synchronization module for the mobile application, allowing field officers in remote areas with zero connectivity to collect data and sync seamlessly once online.
- **API Ecosystem:** Developed high-performance RESTful APIs to bridge communication between mobile field units and the central web dashboard, facilitating real-time data exchange.
- **Data Reporting:** Built a comprehensive reporting engine using Yii2 and PostgreSQL, providing government stakeholders with actionable insights and progress tracking via an interactive dashboard.

### **Project Information Management System (PIMS - MoHFW)**

Role: Solution Architect

Technologies: PHP (Yii2), PostgreSQL, Apache Superset, REST APIs

- **National Infrastructure Monitoring:** Designed the end-to-end architecture for a mission-critical platform used by the Ministry of Health and Family Welfare (MoHFW) to monitor AIIMS construction projects nationwide.
- **Advanced Analytics:** Integrated Apache Superset to deliver high-level executive dashboards, transforming raw construction data into visual KPIs for senior ministry officials.
- **Field-to-Cloud Integration:** Overlooked the development of cross-platform mobile tools with offline data collection capabilities, ensuring project updates were captured accurately from active construction sites.
- **Process Optimization:** Streamlined the monitoring workflow for multi-billion-rupee infrastructure projects, reducing reporting delays and improving project transparency across the department.

### **Laboratory Information Management System (LIMS - NCDC)**

Role: Solution Architect

Technologies: PHP (Yii2), PostgreSQL, REST APIs, IIHP Integration

- **Process Digitization:** Led the digital transformation of laboratory workflows for the National Centre for Disease Control (NCDC), automating test management and result processing.



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- **Interoperability:** Architected RESTful APIs to facilitate secure, real-time data sharing between NCDC labs and external healthcare systems, reducing the turnaround time for critical diagnostic reports.
- **Data Sovereignty:** Ensured robust data management within PostgreSQL to maintain the integrity and confidentiality of national health laboratory data.

### **Point of Entry (POE) Surveillance Application**

Role: Solution Architect

Technologies: PHP (Yii2), PostgreSQL, REST APIs, Mobile Integration

- **Border Health Surveillance:** Developed a mobile-first application for field officers at international airports to conduct real-time health surveillance and questionnaire-based screening.
- **Live Data Exchange:** Implemented high-concurrency APIs to sync field data directly with the IIHP Platform, enabling immediate analysis of potential health threats at entry points.
- **Operational Efficiency:** Replaced manual reporting processes with a streamlined digital interface, significantly increasing the volume and accuracy of surveillance data collected at key hubs.

### **AQI Alert & Escalation System**

Role: Solution Architect

Technologies: PHP (Yii2), PostgreSQL, REST APIs, IIHP Platform Integration

- **Automated Response Engine:** Engineered an intelligent alert system that monitors air quality data and triggers automated escalations to government officials based on critical AQI thresholds.
- **Real-Time Monitoring:** Integrated the application with the IIHP (Integrated Health Information Platform) for live data synchronization, ensuring high-speed notification delivery.
- **Analytical Dashboard:** Developed a centralized command center to track and analyze response rates from various authorities, improving the accountability of environmental interventions.

## **Critical Rivers**

***Senior Software Engineer (June 2023 – October 2023)***

### **Global 3D Manufacturing & E-Commerce Platform (Shapeways, NY)**

Role: Senior Software Engineer / Team Lead

Technologies: PHP, E-commerce Integrations (Shopify/Etsy), API Orchestration

- **Global Platform Maintenance:** Managed and optimized core business operations for a New York-based 3D printing marketplace, facilitating global sales and fulfillment across multiple international regions.
- **Multi-Channel Integration:** Engineered and maintained sophisticated API integrations with Shopify and Etsy, synchronizing high-volume product listings, orders, and inventory data between the Shapeways engine and external storefronts.
- **Technical Leadership:** Led and mentored a junior development team, conducting code reviews and implementing best practices to ensure code quality and system stability.
- **Logistics & Workflow Optimization:** Enhanced the 3D manufacturing workflow by refining back-end logic for order processing, ensuring seamless communication between the digital storefront and physical 3D printing hubs

## **Tata Medical and Diagnostics**

***Senior Software Engineer (Jan 2022 – June 2023)***

### **Projects:**

#### **AMR (Antimicrobial Resistance) Surveillance Platform**

Role: Technical Lead / Architect

Technologies: PHP, MySQL, REST APIs, Bulk Data Processing, Data Analytics

- **Omni-channel Data Ingestion:** Architected a robust data collection engine to aggregate lab test results from hospitals PAN-India via REST APIs, web interfaces, and high-volume Excel bulk uploads.
- **Clinical Decision Support:** Developed a complex analytics module that processed raw diagnostic data to generate trend reports, directly enabling medical experts to formulate National Treatment Guidelines.



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- **Big Data Filtering:** Engineered a multi-layered filtering system to allow medical researchers to query millions of records by geography, pathogen type, and drug resistance patterns with high performance.
- **National Scalability:** Managed the development and maintenance of the platform to ensure 99.9% uptime for participating labs across India.

### **Konnifel (Internship & Academic Portal)**

Role: Team Lead

Technologies: PHP (Yii), Payment Gateway Integration, Recommendation Algorithms

- **Platform Orchestration:** Led a cross-functional development team to build an end-to-end marketplace connecting professors with interns, managing the full SDLC from architecture to deployment.
- **Module Innovation:** Designed and implemented critical modules including an Automated Exam Engine, a Student Recommendation System, and secure Payment Gateway integrations.
- **Engineering Leadership:** Acted as the primary point of contact for technical architecture decisions, mentoring developers, and ensuring modular, maintainable code across the application.

### **Spice Works Ziff Davis**

***Software Engineer SE II (Aug 2021 – Jan 2022)***

#### **Projects:**

##### **Launchpad & Casper**

Role: Full Stack Developer

Technologies: PHP, JavaScript, Desktop App Development, Lead Gen Engines

- **Lead Generation Architecture:** Developed and optimized Launchpad, a high-traffic web application designed to capture, process, and qualify business leads for global B2B marketing campaigns.
- **Identity Verification (Casper):** Orchestrated the development and maintenance of the Casper Desktop Application, a specialized tool used for cross-platform social verification to ensure lead authenticity and data quality.
- **Performance Engineering:** Improved the efficiency of data capture forms and backend processing logic, reducing latency in the lead delivery pipeline to external clients.
- **Cross-Platform Development:** Managed the dual-stack challenge of maintaining a web-based lead engine while ensuring the stability of a desktop-based verification client.

### **Cisco Systems (Velocis Systems Pvt Ltd)**

***Sr. Software Engineer (Nov 2020 – Aug 2021)***

#### **Projects:**

##### **CNAAP (Cisco Network Automation & Analytics Platform)**

Role: Senior Full-Stack Developer

Technologies: PHP, API Development, Network Automation Logic, Analytics Engines

- **Platform Scalability:** Led the full-stack development of new feature modules for the **Cisco Network Automation & Analytics Platform**, enhancing the system's ability to manage large-scale enterprise network infrastructures.
- **API Modernization:** Engineered and optimized RESTful APIs to facilitate seamless communication between automated networking hardware and the central analytics dashboard.
- **System Optimization & Stability:** Performed deep-level debugging and performance tuning of core platform modules, significantly improving system uptime and reliability for enterprise clients.
- **Feature Engineering:** Developed custom analytics components that allowed network administrators to visualize real-time data trends and automate routine network configuration tasks.

### **Digital India Corporation (MeitY)**

***WEB DEVELOPER (Lead Role) (OCT 2015 – NOV 2020)***

#### **Projects:**

##### **IIDS & Krishi Sarathi (National Agro-Advisory Platform)**



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Role: Lead Architect & Full-Stack Developer (Sole Developer) Technologies: PHP, PostgreSQL, REST APIs, UMANG Integration, IVRS

- National Scale Impact: Conceptualized, architected, and independently developed the Interactive Information Dissemination System (IIDS), which was successfully scaled from five states (including Andhra Pradesh & Telangana) to a PAN-India rollout rebranded as "Krishi Sarathi."
- Multi-Channel Delivery: Engineered a complex "Push-Pull" advisory model, enabling real-time communication between millions of farmers and agricultural experts via web, mobile, and toll-free IVRS systems.
- Super-App Integration: Developed and managed the mission-critical APIs for integration with the UMANG Platform, making agro-services accessible to a national audience through the Government of India's single-window mobile app.
- Architecture Ownership: Managed the entire product lifecycle—from database schema design in PostgreSQL to the development of the front-end and back-end modules, ensuring high availability for critical seasonal advisories.

#### **QR3CMS (QR-Coded Check Post Management System)**

Role: Full-Stack Developer Technologies: PHP, QR Integration, IVRS Click-to-Call, Mobile APIs

- Logistics & Governance: Built a web and mobile-based solution for the transport of minerals in Meghalaya, replacing manual tracking with a high-integrity QR-coded verification system.
- Security & Transparency: Designed the system to prevent illegal mineral transport, integrating Click-to-Call IVRS for immediate verification at state check posts.

#### **IPMP (Infra Project Management System)**

Role: Full-Stack Developer Technologies: PHP, API Development, Analytics Dashboard

- High-Level Monitoring: Developed a specialized project tracking tool for the Chief Minister's Office (Meghalaya) to monitor the real-time progress of infrastructure projects.
- Mobile Field Reporting: Engineered APIs for mobile data collection, allowing site engineers to upload progress photos and data directly to the central dashboard for executive review.

### **CDAC – Delhi (via GA Digital)**

**Web Developer (Sep 2013 – Oct 2015)**

#### **Projects:**

##### **CeG (Centre for e-Governance) Web Portal**

Role: Full Stack Developer

Technologies: PHP (Yii), MySQL, Mercurial, Bitbucket

- End-to-End Ownership: Managed the full-stack development and maintenance of the official CeG portal for the Ministry of Electronics and Information Technology (MeitY).
- Security & Hosting: Spearheaded the deployment process, including server hosting and obtaining mandatory security clearances (STQC/Security Audit) for government web standards.
- Community Engagement: Developed interactive modules for citizen engagement, including a national discussion forum, conference registration systems, and automated feedback collection.
- Administrative Control: Engineered a robust administrative backend to manage visit requests, user moderation, and real-time portal updates.

##### **Online Assessment Engine (Quiz Application)**

Role: Full Stack Developer

Technologies: PHP (Yii), MySQL

- Educational Tech: Designed and deployed a web-based quiz application for students visiting the Centre for e-Governance, featuring a real-time results engine.
- CMS Development: Built a custom Content Management System (CMS) for administrators to manage question banks, categorize topics, and analyse visitor performance trends.

### **Stratbeans Consulting**

**Software Engineer (July 2013–Sep 2013)**

#### **Projects:**



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## **ATP (Autumn Test Pad) – LMS & Evaluation Engine**

Role: Backend Developer

Technologies: PHP (Yii), MySQL, Bitbucket

- Enterprise Training Logic: Developed and enhanced a comprehensive Learning Management System (LMS) used for corporate training and employee performance evaluation.
- Role-Based Access Control (RBAC): Implemented complex multi-user workflows for Administrators, Managers, and Users, ensuring secure data silos and specialized reporting for each tier.
- Feature Engineering: Spearheaded the resolution of critical development tickets, focusing on enhancing the evaluation engine's scoring logic and user interface scalability.

## **Vomozo (Educational Portal for CAT Aspirants)**

Role: Backend Developer

Technologies: PHP (Yii), MySQL

- Competitive Exam Architecture: Contributed to the development of a high-traffic web application tailored for MBA (CAT) aspirants, focusing on mock-test modules and performance tracking.
- Product Iteration: Rapidly addressed architectural enhancements and new feature requests to adapt to changing exam patterns and user feedback.

## **I.I.T. Bombay**

**Jr. Software Engineer (Jan 2012 –Nov 2012)**

### **Projects:**

#### **NMEICT (National Mission on Education through ICT)**

Role: Backend Developer

Technologies: Java, JSP, JavaScript, PostgreSQL, Linux

- Algorithmic Achievement: Developed a sophisticated Automated Seat Allotment Module for national-scale workshops. The system successfully managed complex logic to allot seats across 180+ remote centres based on participant preferences and a First-Come-First Serve (FCFS) algorithm.
- Full-Stack Validation: Engineered robust server-side and client-side validation logic using Java and JavaScript, ensuring data integrity for thousands of synchronous and asynchronous student instructions.
- Database Management: Managed and optimized the PostgreSQL database, ensuring high performance for concurrent registration sessions and regular data synchronization across the mission portal.
- National Impact: Contributed to the "Empowerment of Students & Teachers" initiative, a flagship project by the Ministry of Education to digitize higher education across India.



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